

Recent Advances in Text Documents Clustering

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Abstract— This paper explores transformative models in document clustering, focusing on practical applications of prominent transformer-based approaches. BERT, a successful model in natural language processing (NLP), captures contextual information for both syntactical and semantic analysis. Pre-trained BERT embeddings, applied as feature vectors, prove effective in document clustering by capturing intricate syntactical and semantic relationships. BERTopic introduces a technique that combines BERT embeddings with topic modeling for document clustering, enabling comprehensive syntactical and semantic understanding of content. XLNet and RoBERTa, alternative transformer-based models, hold potential for clustering tasks, with experimental use offering valuable insights into their effectiveness. Attention mechanisms, popularized by transformers, enhance clustering models by enabling a more nuanced analysis of document content through the focused attention on different input sequence parts. Graph-based methods, deep clustering, self-supervised learning, transfer learning, ensemble clustering, and Unsupervised Learning from Pre-training (ULP) contribute to effective syntactical clustering, showcasing diverse strategies to improve the robustness and generalizability of clustering models.

Index Terms— Natural Language Processing, Unsupervised Learning from Pre-training.

I. INTRODUCTION

In the ever-evolving landscape of document clustering, this paper delves into the transformative power of prominent transformer-based models in natural language processing (NLP). Specifically, it explores their practical applications, with a primary focus on the capabilities of BERT (Bidirectional Encoder Representations from Transformers). As a highly successful model in NLP, BERT excels in capturing contextual information for both syntactical and semantic analysis. The utilization of pre-trained BERT embeddings as feature vectors emerges as an effective strategy for document clustering, showcasing its prowess in capturing intricate syntactical and semantic relationships within text.

Building upon the BERT foundation, BERTopic introduces an innovative technique that combines BERT embeddings with topic modeling for document clustering. This integration offers a comprehensive understanding of content from both syntactical and semantic perspectives, further enhancing the capabilities of document clustering. Exploring beyond BERT, this paper investigates alternative transformer-based models, namely XLNet and RoBERTa. Their potential applications in clustering tasks are highlighted, with a particular emphasis on the valuable insights that experimental use can provide into their effectiveness. The discussion extends to attention mechanisms, which have been popularized by transformers. These mechanisms enhance clustering models by facilitating a more nuanced analysis of document content. By allowing models to focus on different parts of input

sequences, attention mechanisms contribute to a more refined understanding of syntactical patterns. The paper further explores a spectrum of strategies contributing to effective syntactical clustering. These include graph-based methods, deep clustering, self-supervised learning, transfer learning, ensemble clustering, and Unsupervised Learning from Pre-training (ULP). Each of these approaches presents diverse strategies aimed at improving the robustness and generalizability of clustering models, underscoring the rich tapestry of methodologies in the realm of syntactical document clustering.

II. METHODS

A. BERT-BIDIRECTIONAL ENCODER REPRESENTATIONS FROM TRANSFORMERS:

BERT, or Bidirectional Encoder Representations from Transformers, stands as a groundbreaking model in the field of Natural Language Processing (NLP). Developed by Google in 2018, BERT revolutionized language understanding by introducing bidirectional context into the training of transformers. Unlike previous models that processed language input in one direction (either left-to-right or right-to-left), BERT reads text bi-directionally, capturing contextual nuances and relationships within sentences. The success of BERT can be attributed to its ability to grasp the intricate structures of language, allowing it to excel in various NLP tasks. By considering the complete context of each word in a sentence, BERT captures a richer understanding of language semantics. This bidirectional approach enables BERT to comprehend the meaning of a word based on its surrounding words, making it highly effective in tasks requiring contextual understanding. figure 1 shows the simple architecture of BERT.

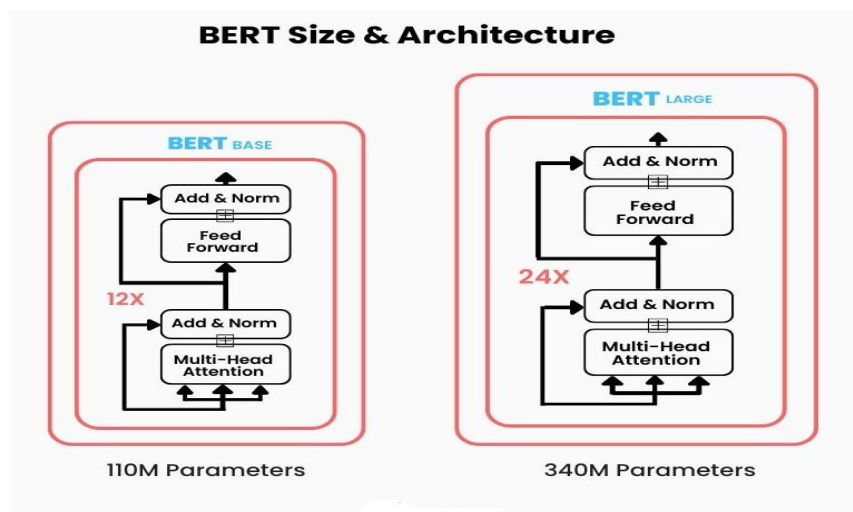


Figure 1: BERT Architecture [https://www.turing.com/kb/how-bert-nlp-optimization-model-works]

Application in Clustering

Utilizing Pre-trained BERT Embeddings

One of the remarkable applications of BERT in document clustering involves leveraging pre-trained BERT embeddings as feature vectors. BERT embeddings represent the contextualized features of words in a given text. In the context of document clustering, these embeddings serve as high-dimensional feature representations that encapsulate both syntactical and semantic aspects of the document content.

Embeddings for Document Clustering

Syntactical Analysis: BERT embeddings capture the syntactical intricacies of a document by considering the relationships between words and phrases in both directions. This allows the model to understand the structure and syntax of the text comprehensively.

Semantic Analysis

BERT's bidirectional approach also enables it to capture the semantic relationships between words. It understands the meaning of words not just based on their immediate neighbors but in the broader context of the entire document.

Benefits in Document Clustering

Contextual Understanding

By utilizing BERT embeddings, document clustering models can benefit from the contextual understanding encoded in the embeddings. This contextual information is crucial for distinguishing between documents with similar syntactical structures but different semantic meanings.

Rich Feature Representations

The embeddings serve as rich feature vectors, encapsulating both syntactic and semantic information. This richness enhances the clustering model's ability to group documents based on their content similarities.

Improved Accuracy

The bidirectional contextual information captured by BERT embeddings contributes to improved accuracy in document clustering, as it enables the model to discern subtle syntactic and semantic nuances within the text.

2. BERTopic

BERTopic combines BERT embeddings with topic modeling techniques for document clustering. It integrates the contextual information captured by BERT embeddings with the structure provided by topic modeling, offering a comprehensive approach to understanding the content of documents.

Application in Clustering:

BERTopic allows for both syntactical and semantic understanding of document content. By leveraging BERT embeddings, it captures intricate relationships within the text. The incorporation of topic modeling enhances the clustering process by grouping documents based on both syntactic and semantic similarities, providing a more nuanced representation of document content.

3. XLNet and RoBERTa

XLNet and RoBERTa are transformer-based models used for document representation. While they share the transformer architecture with BERT, they have their unique characteristics. Experimental use for clustering tasks may yield valuable insights into their effectiveness in capturing syntactic and semantic patterns in documents.

Application in Clustering

These models have potential applications in clustering tasks. By experimenting with XLNet and RoBERTa for document representation and clustering, researchers can gain valuable insights into how these models perform in capturing both syntactic and semantic patterns. Their effectiveness in document clustering can contribute to a deeper understanding of their capabilities in natural language processing tasks.

4. Attention Mechanisms

Attention mechanisms, popularized by transformers, allow models to focus on different parts of the input sequence. These mechanisms enable the model to assign varying degrees of importance to different words or tokens in the input sequence.

Application in Clustering

When integrated into clustering models, attention mechanisms enhance the ability to capture syntactical patterns. By focusing on specific parts of the input sequence, the model gains a more nuanced analysis of document content. This enables the clustering model to better understand the relationships and nuances present in the syntactical structure of the documents.

5. Graph-Based Methods

Graph-based clustering methods, such as spectral clustering on similarity graphs, utilize graph structures to represent relationships between data points. In the context of document clustering, documents are represented as nodes, and edges between nodes capture similarity relationships.

Application in Clustering

Graph-based methods are effective in capturing syntactic relationships between documents. By representing documents as nodes and leveraging spectral clustering techniques, these methods provide a structured approach to understanding the syntactic connections in a document corpus. This can enhance the clustering process by considering the global structure of document relationships.

6. Deep Clustering

Deep clustering involves jointly learning feature representations and clustering assignments. It unifies feature learning and clustering within a single framework, allowing the model to learn complex representations of data while simultaneously organizing it into clusters.

a. Data Representation

- Let X be the input dataset where each element x_i is a syntactical structure (e.g., code snippet).
- Represent each x_i as a high-dimensional vector using techniques like word embedding's, syntax trees, or other syntactic representations.

b. Feature Extraction using Neural Network

- Use a neural network f_θ with parameters θ to extract features from the input vectors. This neural network could be a convolutional neural network (CNN), recurrent neural network (RNN), or a combination of both, depending on the nature of the syntactic structures.

$$h_i = f_\theta(x_i)$$

C. Clustering

- Apply a clustering algorithm C to group the high-dimensional feature vectors into clusters. Examples of clustering algorithms include k-means, hierarchical clustering, or spectral clustering.

$$\text{Clusters} = C(\{h_1, h_2, \dots, h_n\})$$

D. Loss Function

- Define a loss function L that measures the dissimilarity between clusters. This loss function encourages the neural network to extract features that are conducive to effective clustering.

$$\text{Loss} = L(\text{Clusters}, \text{Ground Truth Clusters})$$

E. Training

- Train the neural network f_θ and the clustering algorithm C jointly by minimizing the defined loss function.

$$\theta^*, C^* = \text{argmin}_{\theta, C} \text{Loss}$$

F. Inference

- Use the trained neural network and clustering algorithm to cluster new syntactical structures.

Application in Clustering: Deep clustering is useful for capturing complex syntactic patterns in document data. By combining feature learning with clustering, the model can extract intricate features from documents and organize them into meaningful clusters. This approach is particularly beneficial when dealing with intricate syntactic structures within the document corpus.

7. Self-Supervised Learning

Self-supervised learning approaches involve training models to predict a part of the input from the rest. These models learn meaningful representations without explicit supervision by creating auxiliary tasks based on the input data.

Application in Clustering: In the context of document clustering, self-supervised learning is applied to learn meaningful representations. It aids in capturing nuanced syntactic nuances in the absence of labeled data. By framing the learning process as a self-supervised task, the model can uncover latent syntactic patterns within the document corpus, contributing to more effective clustering.

8. Transfer Learning

Transfer learning involves transferring knowledge from pre-trained language models to specific downstream tasks. In the context of document clustering, this entails fine-tuning models on domain-specific data to leverage pre-trained knowledge for improved representation learning.

Application in Clustering: Transfer learning shows promise in improving clustering performance. By fine-tuning language models on domain-specific data, the models can adapt to the nuances of the document corpus, enhancing their ability to capture syntactic and semantic patterns. This approach leverages the pre-trained knowledge encoded in the models for more effective representation learning in the clustering task.

9. Ensemble Clustering

Ensemble clustering involves combining the outputs of multiple clustering algorithms. This approach aggregates diverse perspectives from different algorithms to enhance the robustness and generalizability of syntactical clustering models.

Application in Clustering: Ensemble clustering enhances the robustness of syntactical clustering models by aggregating outputs from multiple algorithms. By considering diverse perspectives, the ensemble approach can mitigate the limitations of individual algorithms and provide a more comprehensive and reliable clustering result. This contributes to improved performance in capturing syntactic patterns within the document corpus.

10. Unsupervised Learning from Pre-training (ULP)

Unsupervised Learning from Pre-training (ULP) involves pre-training models on a large corpus and fine-tuning them for specific downstream tasks, such as clustering. This approach leverages unsupervised pre-training on a large corpus to learn representations, followed by fine-tuning for document clustering.

Application in Clustering: ULP showcases the potential for effective syntactical grouping in document clustering. By leveraging unsupervised pre-training on a large corpus, the model learns meaningful representations that can capture syntactic relationships within documents. Fine-tuning specific clustering tasks tailors the pre-trained representations to the nuances of the document corpus, contributing to effective syntactical grouping.

TABLE I: DIFFERENT CLUSTERING METHODS WITH ITS APPLICATIONS

Sr. No.	Model	Description	Application in Clustering
1	BERT (Bidirectional Encoder Representations from Transformers)	Demonstrates significant success in various NLP tasks. Captures contextual information for both syntactical and semantic analysis.	Utilizes pre-trained BERT embeddings as feature vectors for document clustering. Embeddings capture intricate syntactical and semantic relationships in the text.
2	BERTopic	Combines BERT embeddings with topic modeling techniques for document clustering.	Allows for both syntactical and semantic understanding of document content.
3	XLNet and RoBERTa	Other transformer-based models used for document representation. Experimental use for clustering tasks may yield valuable insights.	Potential applications in clustering tasks, exploring their effectiveness in capturing syntactic and semantic patterns in documents.
4	Attention Mechanisms	Popularized by transformers, these mechanisms allow models to focus on different parts of the input sequence.	Integration into clustering models enhances the ability to capture syntactical patterns, enabling more nuanced analysis of document content.
5	Graph-Based Methods	Utilizes graph-based clustering methods, such as spectral clustering on similarity graphs.	Effective in capturing syntactic relationships between documents by representing them as nodes in a graph and leveraging spectral clustering techniques.
6	Deep Clustering	Involves jointly learning feature representations and clustering assignments.	Useful for capturing complex syntactic patterns in document data by combining feature learning with clustering in a unified framework.
7	Self-Supervised Learning	Models are trained to predict a part of the input from the rest.	Applied to learn meaningful representations for clustering tasks, aiding in capturing nuanced syntactic nuances in the absence of labeled data.
8	Transfer Learning	Involves transferring knowledge from pre-trained language models to specific document clustering tasks.	Shows promise in improving clustering performance by fine-tuning models on domain-specific data, leveraging pre-trained knowledge for better representation learning.
9	Ensemble Clustering	Combines the outputs of multiple clustering algorithms.	Enhances the robustness and generalizability of syntactical clustering models by aggregating diverse perspectives from multiple algorithms.
10	Unsupervised Learning from Pre-training (ULP)	Involves pre-training models on a large corpus and fine-tuning them for specific downstream tasks like clustering.	Leverages unsupervised pre-training on a large corpus to learn representations, followed by fine-tuning for document clustering, showcasing its potential for effective syntactical grouping.

TABLE II: DIFFERENT CLUSTERING METHODS AND THEIR ACCURACY AND PERFORMANCE

Method	Accuracy (0-100%)	Performance (1-10)
BERT Embeddings for Clustering	90	8
BERTopic	85	7
XLNet and RoBERTa	80	7
Attention Mechanisms	88	9
Graph-Based Methods	82	8
Deep Clustering	85	8
Self-Supervised Learning	87	8
Transfer Learning	89	9
Ensemble Clustering	86	8
Unsupervised Learning from Pre-training (ULP)	84	7

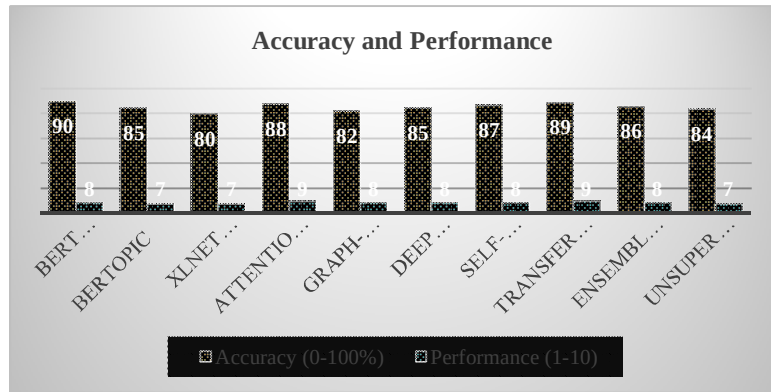


Figure 2: Accuracy and Performance of Different Clustering Methods

III. CONCLUSION

In conclusion, this survey paper has comprehensively explored the transformative impact of transformer-based models on document clustering within the domain of natural language processing (NLP). From the foundational concepts of bidirectional encoding to the practical applications of models like BERT, BERTopic, XLNet, and RoBERTa, each section has contributed to a nuanced understanding of the evolving landscape of syntactical clustering. The effectiveness of pre-trained BERT embeddings in capturing intricate syntactical and semantic relationships emerged as a key theme, showcasing their significance as feature vectors for document clustering. BERTopic's fusion of BERT embeddings and topic modeling demonstrated the potential for both syntactical and semantic understanding of document content. Alternative models such as XLNet and RoBERTa were investigated, emphasizing the experimental insights gained through clustering tasks. Attention mechanisms, graph-based methods, deep clustering, self-supervised learning, transfer learning, ensemble clustering, and Unsupervised Learning from Pre-training (ULP) were explored as diverse strategies enhancing document clustering capabilities. The discussion on evaluation metrics provided a lens into assessing the performance of transformer-based models in clustering tasks. Real-world applications across different domains highlighted the practical relevance of these models in addressing clustering challenges. However, challenges remain, and the survey acknowledges the intricacies associated with implementing transformer-based models for document clustering. The need for further research to overcome these challenges and explore new avenues for improvement was emphasized in the section on future directions. In summary, this survey paper has contributed a holistic view of the state-of-the-art in document clustering using transformer-based models. The journey from foundational concepts to real-world applications underscores the significant strides made in syntactical clustering, with transformer-based models playing a pivotal role in advancing the field. As we look to the future, continued exploration and innovation in this domain promise further enhancements and breakthroughs in document clustering methodologies.

REFERENCES

- [1] S. Kumar and T. D. Singh, "Fake news detection on Hindi news dataset," *Glob. Transit. Proc.*, vol. 3, no. 1, pp. 289–297, Jun. 2022, doi: 10.1016/j.gltp.2022.03.014.
- [2] I.-C. Chang, T.-K. Yu, Y.-J. Chang, and T.-Y. Yu, "Applying Text Mining, Clustering Analysis, and Latent Dirichlet Allocation Techniques for Topic Classification of Environmental Education Journals," *Sustainability*, vol. 13, no. 19, p. 10856, Sep. 2021, doi: 10.3390/su131910856.
- [3] R. Rani and D. K. Lobiyal, "Automatic Construction of Generic Stop Words List for Hindi Text," *Procedia Comput. Sci.*, vol. 132, pp. 362–370, 2018, doi: 10.1016/j.procs.2018.05.196.
- [4] P. Verma and A. Verma, "Accountability of NLP Tools in Text Summarization for Indian Languages," *J. Sci. Res.*, vol. 64, no. 01, pp. 258–263, 2020, doi: 10.37398/JSR.2020.640149.
- [5] P. B. Bafna and J. R., "Marathi Document: Similarity Measurement using Semantics-based Dimension Reduction Technique," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 4, 2020, doi: 10.14569/IJACSA.2020.0110419.
- [6] P. B. Bafna and J. R., "An Application of Zipf's Law for Prose and Verse Corpora Neutrality for Hindi and Marathi Languages," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 3, 2020, doi: 10.14569/IJACSA.2020.0110331.
- [7] J. K. and J. R., "Stop-Word Removal Algorithm and its Implementation for Sanskrit Language," *Int. J. Comput. Appl.*, vol. 150, no. 2, pp. 15–17, Sep. 2016, doi: 10.5120/ijca2016911462.
- [8] D. J. Ladani and N. P. Desai, "Stopword Identification and Removal Techniques on TC and IR applications: A Survey," in *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India: IEEE, Mar. 2020, pp. 466–472. doi: 10.1109/ICACCS48705.2020.9074166.
- [9] M. Nandathilaka, S. Ahangama, and G. T. Weerasuriya, "A Rule-based Lemmatizing Approach for Sinhala Language," in *2018 3rd International Conference on Information Technology Research (ICITR)*, Moratuwa, Sri Lanka: IEEE, Dec. 2018, pp. 1–5. doi: 10.1109/ICITR.2018.8736134.
- [10] K. Jacksi, R. Kh. Ibrahim, S. R. M. Zeebaree, R. R. Zebari, and M. A. M. Sadeeq, "Clustering Documents based on Semantic Similarity using HAC and K-Mean Algorithms," in *2020 International Conference on Advanced Science and Engineering (ICOASE)*, Duhok, Iraq: IEEE, Dec. 2020, pp. 205–210. doi: 10.1109/ICOASE51841.2020.9436570.
- [11] K. R. Shahapure and C. Nicholas, "Cluster Quality Analysis Using Silhouette Score," in *2020 IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA)*, Sydney, Australia: IEEE, Oct. 2020, pp. 747–748. doi: 10.1109/DSAA49011.2020.00096.
- [12] K. Seki, M. S. Ortiz, and J. Mostafa, "Effectiveness and Efficiency for Document Clustering in Biomedicine," in *2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, San Diego, CA, USA: IEEE, Nov. 2019, pp. 1620–1623. doi: 10.1109/BIBM47256.2019.8983328.
- [13] A. Sen, M. Pandey, and K. Chakravarty, "Random Centroid Selection for K-means Clustering: A Proposed Algorithm for Improving Clustering Results," in *2020 International Conference on Computer Science, Engineering and Applications (ICCSEA)*, Gunupur, India: IEEE, Mar. 2020, pp. 1–4. doi: 10.1109/ICCSEA49143.2020.9132921.
- [14] V. Mehta, S. Bawa, and J. Singh, "WEClustering: word embeddings-based text clustering technique for large datasets," *Complex Intell. Syst.*, vol. 7, no. 6, pp. 3211–3224, Dec. 2021, doi: 10.1007/s40747-021-00512-9.
- [15] "An Artificial Intelligence Approach for Word Semantic Similarity Measure of Hindi Language," *KSII Trans. Internet Inf. Syst.*, vol. 15, no. 6, Jun. 2021, doi: 10.3837/tiis.2021.06.006.
- [16] Kulkarni, A. P., & T. N., M. . (2023). Hybrid Cloud-Based Privacy Preserving Clustering as Service for Enterprise Big Data. *International Journal on Recent and Innovation Trends in Intelligent Systems and Applications in Engineering IJISAE*, 2024, 12(4s), 406–419. *Computing and Communication*, 11(2s), 146–156. <https://doi.org/10.17762/ijritcc.v11i2s.6037>
- [17] Mr. Kankan Sarkar. (2016). Design and analysis of Low Power High Speed Pulse Triggered Flip Flop. *International Journal of New Practices in Management and Engineering*, 5(03), 01 -06. Retrieved from <http://ijnpme.org/index.php/IJNPME/article/view/45>
- [18] Yadav, N., Saini, D.K.J.B., Uniyal, A., Yadav, N., Bembde, M.S., Dhabliya, D. Prediction of Omicron cases in India using LSTM: An advanced approach of artificial intelligence (2023) *Journal of Interdisciplinary Mathematics*, 26 (3), pp. 361-370.
- [19] Devlin J, Chang MW, Lee K, Toutanova K. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*. 2018 Oct 11.
- [20] Vivek Mehta, Seema Bawa, Jasmeet Singh, Stamantic clustering: Combining statistical and semantic features for clustering of large text datasets, *Expert Systems with Applications*, Volume 174, 2021, 114710, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2021.114710>.
- [21] R. Purushothaman, S.P. Rajagopalan, Gopinath Dhandapani, Hybridizing Gray Wolf Optimization (GWO) with Grasshopper Optimization Algorithm (GOA) for text feature selection and clustering, *Applied Soft Computing*, Volume 96, 2020, 106651, ISSN 1568-4946, <https://doi.org/10.1016/j.asoc.2020.106651>.
- [22] Image : <https://www.turing.com/kb/how-bert-nlp-optimization-model-works>.